

Ans ①: Text Generation techniques

① Autoregressive models:

- Generate one token at a time, predicting each next words based on prior context.
eg. "I love pizza" → predicts "and" → "I love Pizza and"
- This approach underlies models like GPT-2, GPT-3, and ChatGPT.

Sampling strategies.

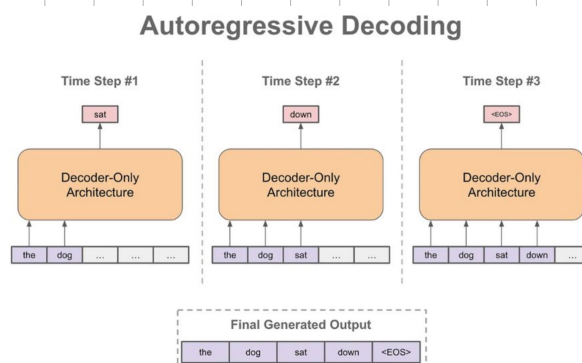
These control how the next tokens is chosen.

Strategy	Behaviour	Outcome
Greedy	Always pick the most likely token	Risky: repetitive outputs.
Top k	Choose randomly from top k tokens	More diverse output.
Top P	Sample from smallest set whose cumulative probability $\text{prob} \geq p$ (eg 0.9)	Balanced and dynamic
Temperature	Scale probability to increase/decrease randomness	Controls creativity level.

→ Top-p (nucleus sampling) adapt the size of candidate pool based on probability mass - typically preferable over fixed k values.

→ These methods are implemented in frameworks like Hugging face Transformers.

→ The diagramme visualise how probability distributions and branching decision shape generated sequences.



Ans ② Building Conversational AI.

Conversational Systems (chatbots, virtual assistance) typically include.

② Comp Elements

- NLU (Natural language understanding): Identify user intent and extract entity.
- Dialogue Manager: Maintain conversation context and flow.
- Backend/knowledge layer: Reference facts or trigger actions.

④ Conversational flow:

- ① User speaks or types
- ② NLU extracts meaning and intent
- ③ Dialogue manager select next steps or retrieve info.
- ④ Response generator creates and sends a reply.

③ Tools:

- Rasa, Dialogflow: Ideal for rule based + intent-based chatbots.
- Hugging face and OpenAI API's: Excellent for model based dialogue generation.
- Additional deployment options include platform like AWS, Azure, or enterprise frameworks.

Ques ③: Ethical Consideration in AI Generated texts.

Ans: ① Bias:

- AI model learns from human-generated data. They inherit social, gender, and racial biases.
- Example. Associating certain professions with specific genders.
- Mitigation:
 - Balanced dataset
 - Bias detection and filtering.

③ Misinformation:

- AI models can generate plausible but false content.
- Risks: Fake news, deep fakes, or misleading narratives.
- Mitigation:
 - Fact checking modules
 - Human moderation
 - Traceability and source verification

④ Responsible AI

- Transparency: Disclose when users interact with AI.
- Accountability: Clear ownership of output
- Privacy: Avoid storing sensitive user data.
- Fairness: Ensure no discrimination in model outputs.
- Safety: Prevents harmful or toxic response.

Ques ④: Generative AI models.

Ans ④: Generative AI refers to models that create new data (text, images, music, video, code etc) that resembles real world data.

Types of Generative data.

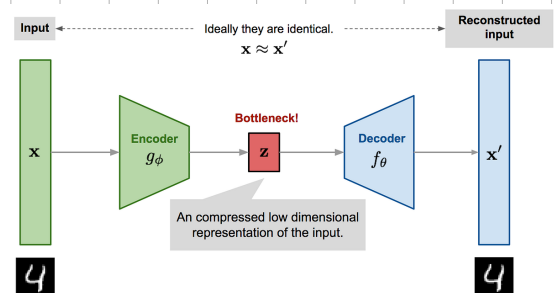
① Autoregressive models:

- Generates data step by step, each output depending on previous outputs.
- Used mostly in text generation.
- Examples are GPT, GPT2 ..., LLaMA, PaLM
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$$P(x) = P(x_1) + P(x_2|x_1) + P(x_3|x_1, x_2) \dots$$

② Variational autoencoders (VAE's)

- Architecture:
- Generates new data similar to input data.
- Encoder: Converts data into latent representation (compressed form).



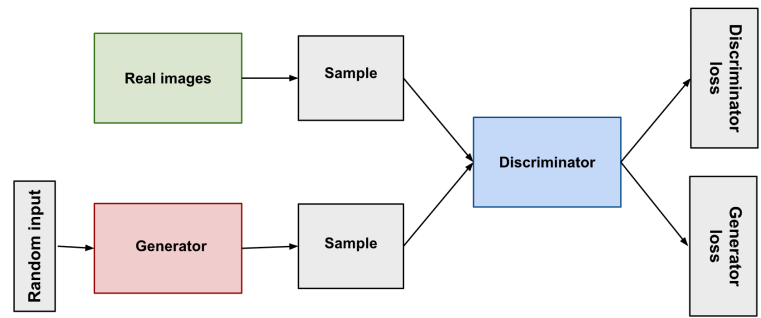
decoder: Reconstructs new data from latent space.

- It is used for image generation, anomaly detection, data compression
- Example: Generating handwritten digits similar to MNIST.

③ GAN: Generative adversarial Neural network.

- Two models competing with each other
- Generator: Creates fake samples.

Discriminator: Detects whether samples are real and or fake.



- Generator learns to fool the discriminator
- Applications: Image synthesis (faces, art)
Deepfake generation